

GTIIT Spring 2025 Calculus 1M

Workshop 14: Exam preparatory

Important note: This list **does not** reflect the structure or content of Exams A and B. It is only a practice list intended to briefly review the course material.

To be fully prepared, you should review all the course content and solve many exercises from the main bibliography.

Exercise 1. Test the convergence of the improper integral

$$\int_1^{\infty} \frac{\sqrt{x}}{x^3 + 1} dx,$$

and compute its value if it converges.

Exercise 2. Determine whether the series

$$\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{3^n}$$

converges or diverges, and justify your answer.

Exercise 3. Find the radius and interval of convergence of the power series

$$\sum_{n=0}^{\infty} \frac{(2n)!}{n!} x^n.$$

Exercise 4. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with $f(1) = 2$, $f'(1) = -1$, $g(1) = 1$, $g'(1) = 3$. Define $h(x) = f(g(x^2))$. Find the equation of the tangent line to $y = h(x)$ at $x = 1$.

Exercise 5. Determine whether the series

$$\sum_{n=1}^{\infty} \frac{n!}{2^n}$$

converges or diverges.

Exercise 6. Compute

$$\lim_{x \rightarrow 0} \frac{\tan(3x) - 3x}{x^3}.$$

Exercise 7. Solve for a, b such that the improper integral

$$\int_0^{\infty} \frac{x^a}{(1+x^b)^2} dx$$

converges.

Exercise 8. Determine the convergence or divergence of the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\ln n}{n}.$$

Exercise 9. Evaluate the limit

$$\lim_{x \rightarrow \infty} \left(\frac{x+1}{x+4} \right)^{2x+3}.$$

Exercise 10. Let

$$f(x) = \begin{cases} Ax^3 + B, & x \leq 2, \\ \ln(x+1) + C, & x > 2. \end{cases}$$

Find the real constants A , B , and C so that f is continuous and differentiable at $x = 2$.

Exercise 11. Find all global minima and maxima of the function

$$g(x) = \frac{x^4 - x^3 - 8}{x^2 - x - 6} \quad \text{on } \mathbb{R},$$

and classify each as local or global.

Exercise 12. Determine whether each improper integral converges or diverges. If it converges, find its value:

(a) $\int_1^{\infty} \frac{\ln x}{x^2} dx.$

(b) $\int_0^{\pi/2} \tan^p x \, dx$, where $p > 1$.

Exercise 13. Compute the following one-sided limits:

$$\lim_{x \rightarrow 0^+} \frac{1 - \cos x}{x^2}, \quad \lim_{x \rightarrow 0^-} \frac{e^x - 1}{x}.$$

Exercise 14. Differentiate implicitly to find $\frac{dy}{dx}$ for

$$x^2 + xy + y^3 = \sin(xy).$$

Exercise 15. Let

$$F(x) = \int_{\ln 2}^{x^2} \frac{t^2}{\sqrt{1+t^3}} dt.$$

Find $F'(x)$ and evaluate $F'(\sqrt{3})$.

Exercise 16. Perform a partial-fraction decomposition of

$$\frac{3x+5}{(x-1)(x+2)^2}$$

and hence compute $\int \frac{3x+5}{(x-1)(x+2)^2} dx.$